

INTERNSHIP PRESENTATION

Frontend Web Development Internship

Engineering Responsive Interfaces

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B.Tech CSE (Batch 2025-2029) | IILM University

INTERNSHIP OVERVIEW

Bridging Academia and Industry

Practical Web Architecture

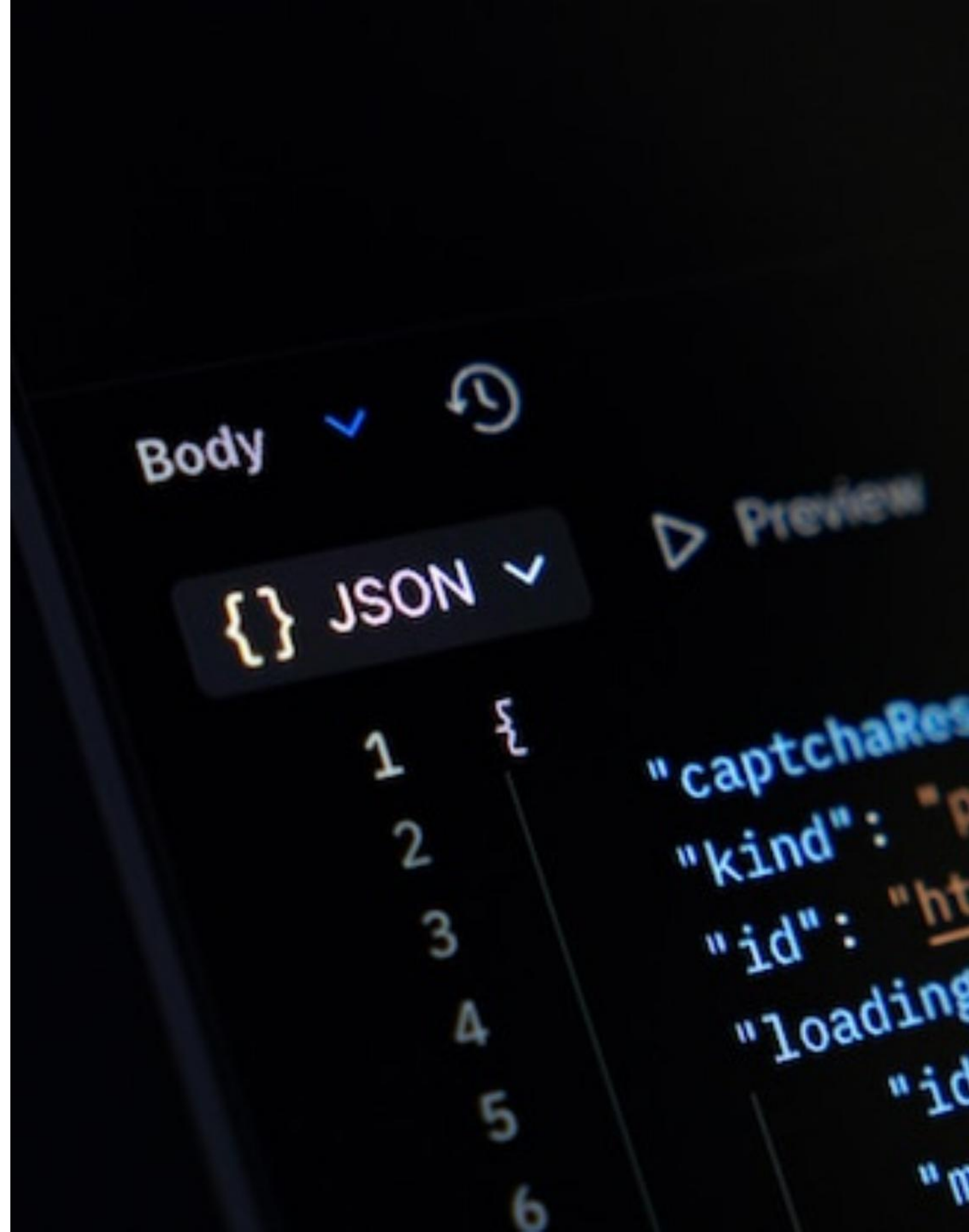
Translating theoretical principles into industry-standard engineering.

Framework-Free Deliverables

Independent design using strictly native web technologies.

Production Sprints

Dynamic image gallery and vintage calculator applications.





ORGANIZATION PROFILE

Code Alpha

Closing the gap between university curricula and practical software industry demands.

Virtual Project Sprints

Realistic tasks replicating standard team deliverables.

Clean Architecture

Foundational web structure and strict separation of concerns.

Strict Version Control

Robust repository management via Git and GitHub.

Core Technical Objectives



The Technical Challenge

High-performance native vanilla interfaces.



Image Gallery

Fluid grids with synchronized modal navigation.



Vintage Calculator

Skeuomorphic interactions with fault-proof logic.

System Architecture

01

Semantic HTML5

Accessible content architecture.

02

Advanced CSS3

Responsive multi-column grids.

03

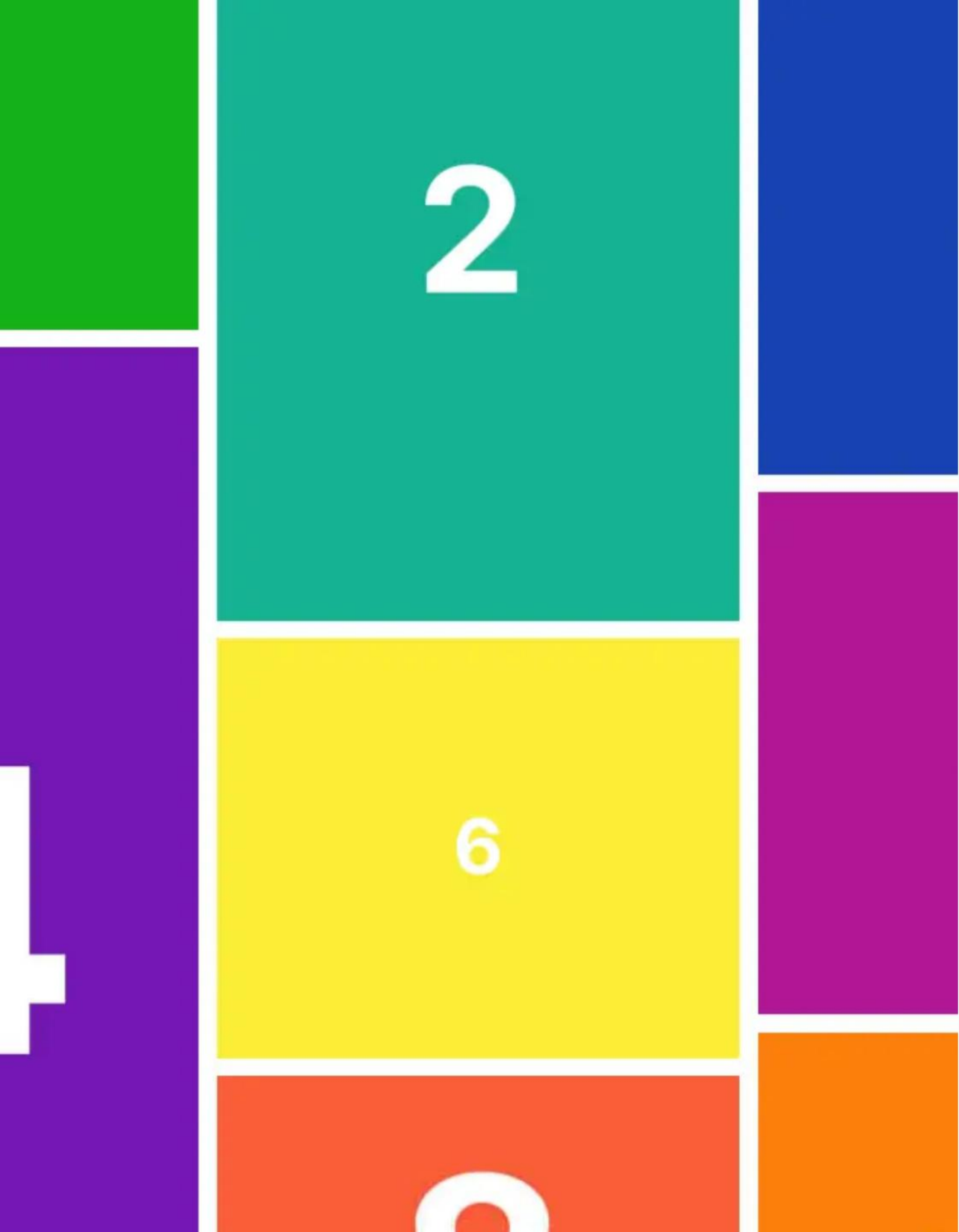
Vanilla JS (ES6+)

Primary application logic engine.

04

MVC Pattern

Strict presentation layer separation.

A decorative background on the left side of the slide consists of several colored squares and rectangles. A large teal square contains the number '2'. Below it, a yellow square contains the number '6'. At the bottom, an orange square contains the number '9'. To the right of the teal square is a blue rectangle. To the right of the yellow square is a purple rectangle. To the right of the orange square is an orange rectangle. On the far left, a purple rectangle contains a white plus sign. The numbers '2', '6', and '9' are white and centered within their respective squares.

2

PROJECT 01

Interactive Image Gallery

Fluid Responsive Layout

Auto-fitting CSS grids scaling across all viewports.

Dynamic Lightbox Modal

Full desktop keyboard navigation and dismissal.

Asynchronous Filtering Engine

Synchronized index tracking via dynamic visibility toggles.

PROJECT 02

Retro-Themed Calculator

Skeuomorphic UI/UX

Authentic tactile 3D key depression effects.

Fault-Tolerant Engine

Safe sequential operation chaining without zero-division crashes.

Keyboard Integration

Direct mapped physical numpad listeners.



Expected Outcomes & Learnings

Production Ready

Polished web applications reflecting professional standards.

Mechanic Mastery

Deep competence in JavaScript event loops and layouts.

Strong Foundation

Highly stable architecture for future complex applications.

Image Sources



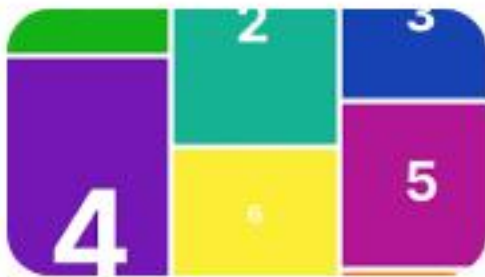
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